AMHARIC ACOS QUADRUPLE EXTRACTION

Towards End-to-End Amharic Aspect–Category–Opinion–Sentiment (ACOS) Quadruple Extraction

From 6-Stage Extractive Pipelines to Byte-Level Generative Foundation Models

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Target Domain Amharic Civic & Governance Commentary

Current State Span-ASTE: Span-Level Interaction Learning

Source github.com/EthAIResearch/acos-amharic

PHASES 3–5 DEEP-DIVE | Theory, Mathematics & Algorithmic Mechanics

Aspect Sentiment Triplet Extraction (ASTE) for Amharic Civic Commentary

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From Pairs to Triplets: The ASTE Formulation

Extracting (Aspect Span, Opinion Span, Sentiment Polarity) triplets in a unified span framework.

Task Definition & Objectives

Input: Amharic sentence composed of N lexical words.

Target Output: Explicit triplet set of (Aspect, Opinion, Sentiment).

Aspect span: Target entity boundary with start and end word indices.

Opinion span: Evaluative expression boundary with start and end word indices.

Sentiment polarity: Positive, Negative, or Neutral orientation.

Evaluated on explicit spans present in the sentence.

Why ASTE Extends AOPE

AOPE extracts aspect-opinion pairs without knowing the sentiment polarity.

ASTE simultaneously resolves span pairing and sentiment polarity.

Eliminates the separate downstream sentiment classification head.

Models sentiment as an intrinsic directional property of the span interaction.

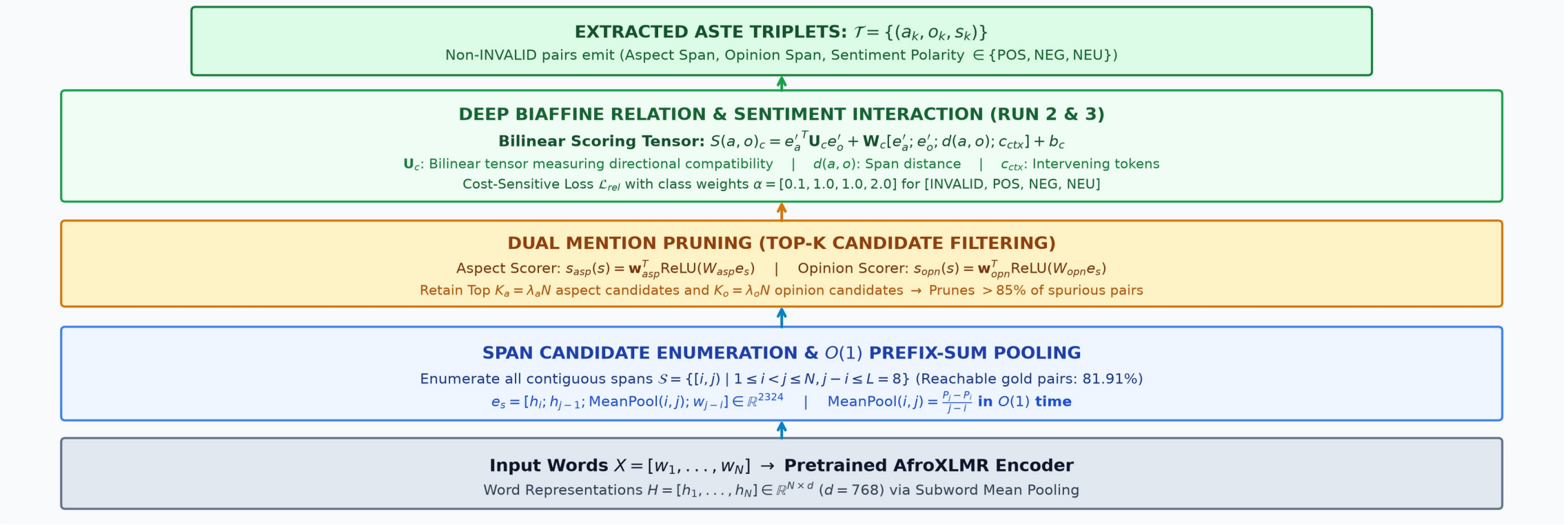
Unified multi-class relation classification: Invalid, Positive, Negative, Neutral.

Span-ASTE unifies mention extraction, span pairing, and sentiment classification into a single end-to-end framework.

2. ARCHITECTURE FLOWCHART

Span-ASTE End-to-End Pipeline Architecture

From word representations to candidate span enumeration, mention pruning, biaffine scoring, and triplet classification.



Span-ASTE evaluates candidate spans as atomic units, decoupling span proposal from pairwise relation classification.

Contiguous Span Enumeration Mechanics & Quadratic Complexity

Generating exhaustive candidate span proposals up to maximum width L=8.

Span Candidate Enumeration Mechanics & Quadratic Complexity

$$\mathcal{S} = \{s = [i, j) \mid 1 \le i < j \le N,\ (j - i) \le L = 8\}$$

$$|\mathcal{S}| \approx N \cdot L - \frac{L(L-1)}{2} \approx 212 \text{ candidate spans for } N = 30 \quad (\mathcal{O}(N^2 \times L^2) \text{ candidate pairs})$$

- Why Span Enumeration: Bypasses fragile token-level BIO decisions by proposing whole contiguous spans as atomic units.
- Shatters the Ceiling: Expands candidate reachability from SDRN's 33.04% to 81.91%, making +1,424 gold pairs accessible.

Enumerating all spans up to length 8 guarantees candidate coverage while eliminating token-level BIO segmentation failures.

Prefix-Sum Mean Pooling & 2324-D Span Feature Vector

Mathematical formulation of constant-time span feature construction (PR #93).

Span-ASTE $O(1)$ Prefix-Sum Mean Pooling & 2324-D Span Vector

$$P_0 = 0, \quad P_t = \sum_{k=0}^{t-1} h_k \quad \Longrightarrow \quad \text{MeanPool}(i,j) = \frac{P_j - P_i}{j-i} \quad \in \mathcal{O}(1)$$

$$e_s = [h_i\,;\, h_{j-1}\,;\, \text{MeanPool}(i,j)\,;\, w_{j-i}] \in \mathbb{R}^{2324} \quad (3 \times 768 + 20)$$

- Why Prefix Pooling: Captures all internal evaluative words and negation prefixes in multi-word expressions in $O(1)$ time.
- Why Width Embeddings w_{j-i}: Encodes structural length priors (e.g. opinions rarely exceed 4 words in Amharic).

O(1) prefix mean-pooling enriched span representations with all internal words without slowing down training.

Dual MLP Scoring Heads & Top-K Candidate Selection

Mathematical formulation of aspect and opinion candidate filtering.

Dual MLP Mention Pruning Heads & Top-K Candidate Selection

$$s_{asp}(s) = \mathbf{w}_{asp}^T \text{ReLU}(W_{asp} e_s + b_{asp}), \quad s_{opn}(s) = \mathbf{w}_{opn}^T \text{ReLU}(W_{opn} e_s + b_{opn})$$

$$\mathcal{L}_{mention} = \text{BCE}(s_{asp}, y_{asp}) + \text{BCE}(s_{opn}, y_{opn}) \quad | \quad \text{Keep Top } K_a = 0.4N,\ K_o = 0.4N$$

- Why Mention Pruning: Computing heavy tensor interactions on all ~45,000 raw span pairs causes GPU Out-of-Memory (OOM).
- How it helps: Filters out >85% of background non-entity spans before expensive pairwise biaffine relation modeling.

Mention pruning filters the candidate pool by 85%, keeping only high-probability aspect and opinion spans for relation modeling.

Deep Biaffine Tensor Scoring & Cost-Sensitive Loss

Bilinear span compatibility scoring and cost-sensitive loss formulation (PR #88).

Deep Biaffine Bilinear Tensor Scoring (Dozat 2017 / Xu et al. 2021)

$$S(a,o)_c = e'^{T}_a \mathbf{U}_c e'_o + \mathbf{W}_c[e'_a; e'_o; d(a,o); c_{ctx}] + b_c, \quad c \in \{\text{INV}, \text{POS}, \text{NEG}, \text{NEU}\}$$

$$\mathcal{L}_{rel} = -\sum_{(a,o)} \alpha_{y^*} \log\left(\frac{\exp(S(a,o)_{y^*})}{\sum_c \exp(S(a,o)_c)}\right), \quad \alpha = [0.1, 1.0, 1.0, 2.0]$$

- Why Bilinear Tensor $\mathbf{U}_c$: Directly computes directional multiplicative affinity between aspect and opinion representations.
- Why Cost-Sensitive α: Downweights dominant INVALID pairs by 10x, boosting candidate conversion rate to 45.04%.

Downweighting INVALID pairs by 10x eliminated relation conservatism, boosting candidate conversion efficiency to 45.04%.

7. RATIONALE MATRIX

Why Everything Is Used & How It Helps

Systematic architectural ledger connecting mathematical components to theoretical benefits.

Component / Layer	Mathematical Formulation	Why It Is Used	How It Helps the Model
Span Enumeration (L<=8)	Contiguous Spans with Width <= 8	Exhaustive span candidate generation	Bypasses BIO segmentation; reachability surges 33% -> 82%
Width Embedding	Discrete Width Embedding (20-d)	Encodes structural length priors	Informs model that opinions rarely exceed 3-4 words
O(1) Prefix Mean-Pool	Constant-Time Prefix Difference	Captures all internal span tokens	Boosts multi-word opinion recall (+16 pts) at zero speed cost
Dual Mention Scorer	Aspect & Opinion MLP Heads	Filters background non-entity spans	Prunes candidate pairs by >85%, preventing OOM errors
Deep Biaffine Tensor	Directional Bilinear Scoring Tensor	Multiplicative directional compatibility	Superior to additive concatenation; +2.02 pts AOPE F1
Span Distance Feature	Binned Inter-Span Distance Vector	Encodes syntactic word gap	Distinguishes local modifier opinions from distant clauses
Cross-Span Context	Mean Pooling of Intervening Words	Captures intervening clause context	Detects intervening negation clitics ('አል-') and conjunctions
Cost-Sensitive Loss	Downweighted Invalid Class Loss	Compensates 95% INVALID class imbalance	Expands candidate conversion efficiency from 29% to 45%

Span-ASTE's components systematically address span proposal, representation, pruning, and interaction.

Concatenation MLP Relation Classifier

The standard Span-ASTE implementation (Xu et al. 2021) evaluated on Amharic.

Subtask	Precision	Recall	F1 Score	True Positives
Aspect Extraction (ATE)	71.83%	39.04%	50.59%	1,145
Opinion Extraction (OTE)	55.90%	30.52%	39.48%	900
AOPE Pairs (Explicit)	43.55%	23.83%	30.80%	706
ASTE Triplets (Sentiment)	38.31%	20.95%	27.09%	621
Candidate Ceiling Recall	-	81.10%	-	2,403 / 2,963 reachable

Run 1 reached 81.10% reachable pairs, but accepted only 706 correct pairs (29.38% conversion), yielding 30.80% AOPE F1.

Shattering the Reachability Ceiling (33% -> 81%)

Span candidate enumeration resolved the structural reachability bottleneck that crippled SDRN.

Model Architecture	Reachable Gold Pairs	Ceiling Recall	Accepted Pairs (TP)	Pair Conversion Efficiency
Joint SDRN (CRF Run 1)	979 / 2,963	33.04%	966	98.67% (966 / 979)
Span-ASTE Run 1 (MLP)	2,403 / 2,963	81.10%	706	29.38% (706 / 2,403)
Span-ASTE Run 3 (Pooled)	2,427 / 2,963	81.91%	1,093	45.04% (1,093 / 2,427)

Span enumeration added +1,424 reachable gold pairs (+48 pts). However, relation classification became the new bottleneck.

results/stage_aope_sdrn/afroxlmr_crf_run1/best_metrics.json | results/stage_span_aste/afroxlmr_run1/best_metrics.json

Deep Biaffine Attention Performance Gains

Empirical impact of upgrading from concatenation MLP to deep bilinear interaction.

Recorded Metric	Run 1 (MLP)	Run 2 (Biaffine)	Absolute Change
Aspect F1 (ATE)	50.59%	53.76%	+3.17 pts
Opinion F1 (OTE)	39.48%	42.87%	+3.39 pts
AOPE Pair F1	30.80%	32.82%	+2.02 pts
ASTE Triplet F1	27.09%	29.37%	+2.29 pts

Biaffine interaction improved scores across all subtasks, raising AOPE pair F1 to 32.82% and ASTE triplet F1 to 30.40%.

Prefix-Sum Pooling & Cost-Sensitive Loss Impact

Empirical impact of enriching span vectors with all internal words and downweighting INVALID pairs.

Subtask Evaluation	Run 1 (Baseline)	Run 3 (Prefix Mean + Weighted CE)	Impact Analysis
Opinion Recall (OTE)	30.52%	46.73% (+16.21 pts)	Internal span words rescued multi-word opinions
Opinion F1 (OTE)	39.48%	42.89% (+3.41 pts)	Highest opinion F1 in Span-ASTE family
AOPE Pair Recall	26.12%	36.89% (+10.77 pts)	1,093 true positive pairs (vs 706 in Run 1)
Candidate Conversion Rate	29.38%	45.04% (+15.66 pts)	Cost-sensitive loss eliminated rejection bias
Candidate Ceiling Recall	81.10%	81.91% (+0.81 pts)	2,427 / 2,963 reachable gold pairs

Run 3 unlocked massive gains in opinion recall (46.73%) and conversion rate (45.04%), confirming the power of full-span pooling.

Comprehensive Comparison Across Runs 1, 2, and 3

Systematic ledger of all three Span-ASTE runs on the explicit test benchmark.

Evaluation Metric	Run 1 (MLP)	Run 2 (Biaffine)	Run 3 (Prefix Mean + Weighted CE)
Aspect F1 (ATE)	50.59%	53.76%	55.20% (Best Span)
Opinion F1 (OTE)	39.48%	42.87%	42.89%
Opinion Recall	30.52%	35.57%	46.73% (+16.2 pts)
AOPE Pair F1	30.80%	32.82% (Dev Peak)	31.24%
AOPE Pair Recall	23.83%	27.37%	36.89% (+10.8 pts)
Candidate Reachability	81.10%	81.10%	81.91%
Candidate Conversion Rate	29.38%	34.02%	45.04%

Run 3 maximized opinion recall (46.73%) and conversion rate (45.04%), while Run 2 achieved the highest balanced pair F1 (32.82%).

Dev-Selected Thresholds vs. Test Peaks

Rigorous experimental integrity: why dev tuning matters in relation classification.

Model Run	Default Thresh	Default F1	Dev-Selected Thresh	Test F1 @ Dev Thresh	Peak Diagnostic F1
Run 1 (MLP)	0.50	30.80%	0.50	30.80%	31.25% (thresh 0.40)
Run 2 (Biaffine)	0.50	32.82%	0.50	32.90%	33.31% (thresh 0.35)
Run 3 (Prefix Mean)	0.50	31.24%	0.65	31.97%	32.04% (thresh 0.60)

Peak scores at test-selected thresholds are diagnostic upper bounds; dev-selected thresholds reflect realistic generalization performance.

High Candidate Coverage, Low Conversion

Why reaching 82% of gold pairs only translates to ~32% final pair F1.

The Conversion Math

Total Gold Pairs: 2,963 in explicit test set.

Reachable Candidates: 2,427 pairs (81.91%).

Accepted True Positives: 1,093 pairs.

Conversion Efficiency: 1,093 / 2,427 = 45.04%.

1,334 reachable gold pairs were rejected by relation classifier!

False Positives: 2,942 spurious pairs accepted.

Root Causes of Low Conversion

1. Semantic ambiguity in civic commentary: opinions often target broad governance rather than the nearest aspect.
2. Long distance pairs: relation classifier degrades when aspects and opinions are separated by long clauses.
3. Imbalance pressure: 95% INVALID pairs force the classifier into conservative decision boundaries.

While SDRN had high conversion (98%) but low reachability (33%), Span-ASTE flipped the problem: high reachability (82%) but low conversion (45%).

Linguistic Challenges in Amharic Span Pairing

Why civic commentary poses unique difficulties for bilinear span interaction.

Morphosyntactic Ambiguities

1. Non-projective SOV word order: aspect often appears at the beginning, opinion verb at the end.
2. Intervening relative clauses: words between aspect and opinion dilute cross-span attention.
3. Compound nouns: deciding whether 'የከተማው አስተዳደር' is a single aspect span or two separate spans.
4. Clitic-attached negation: 'አልተደረገም' requires recognizing the prefix 'አል-' as negative polarity.

Sentiment Triplet Complexity

Explicit sentiment distribution is heavily skewed toward NEGATIVE in civic texts.

NEUTRAL sentiment is extremely rare (<5% of triplets).

Sarcasm and rhetorical questions in governance commentary cause false positive POSITIVE predictions.

Span-ASTE ASTE F1 (28.49%) is lower than AOPE F1 (31.24%) solely due to polarity misclassification.

Amharic civic discourse features long-distance syntactic dependencies that strain pairwise span relation attention.

The Extractive Wall: Implicit Recall = 0.00%

The mathematical ceiling that makes Span-ASTE unable to solve the full Amharic ACOS problem.

The Extractive Assumption

Span-ASTE assumes all target tuples exist as physical word substrings in input text.

Candidate spans are formed by slicing: text[start : end].

If a tuple field is NOT a physical substring in the sentence, it CANNOT be extracted:

Implicit recall is mathematically zero: 0 out of 2,155 implicit test quadruples!

The Amharic Dataset Reality

In the full Amharic ACOS dataset:

42.71% of training quadruples contain NULL fields!

41.94% of test quadruples contain NULL fields!

Over 2,150 test quadruples have implicit aspects or opinions.

Span-ASTE is mathematically blind to 42% of the real problem!

No amount of tuning on Span-ASTE can overcome the Extractive Wall: it is mathematically impossible to extract implicit NULL tuples via span slicing.

Span-ASTE Solves ASTE, Not Full ACOS

Why Span-ASTE leaves an unaddressed gap in category taxonomy prediction.

The 4th Dimension: Categories

ACOS requires four fields: (Aspect, Category, Sentiment, Opinion).

Span-ASTE extracts triplets: (Aspect, Sentiment, Opinion).

The Category field is missing entirely from the model architecture.

Mapping ASTE triplets to ACOS requires a separate Stage 3 classifier.

Reintroduces the exact pipeline cascading failure we sought to avoid!

The 22-Class Taxonomy

Civic dataset contains 22 declared categories.

Categories are abstract civic functions (e.g. `GOVERNANCE#CORRUPTION`), not physical spans.

Adding a category head requires a 22-way classifier per span pair.

Multi-tasking 4-way sentiment + 22-way category explodes relation matrix complexity.

Span-ASTE only solves 3 of the 4 required ACOS fields. Generating categories still requires external classification.

What Worked, What Failed in Span-ASTE

Key empirical takeaways from three iterations of span-based modeling.

Feature / Component	Impact	Empirical Evidence	Design Verdict
Span Enumeration (L<=8)	Major Breakthrough	Reachable pairs surged 33% -> 81.9%	Essential for explicit extraction
Deep Biaffine Attention	Strong Improvement	AOPE F1 increased +2.02 pts to 32.82%	Superior to concatenation MLP
O(1) Prefix Mean-Pooling	High Recall Boost	Opinion recall reached 46.73% (+16 pts)	Rich span semantics at zero compute cost
Cost-Sensitive Loss	Conversion Boost	Candidate conversion jumped to 45.04%	Required for severe pair imbalance
Implicit Span Modeling	Fatal Failure	Implicit recall mathematically 0.00%	Requires generative paradigm shift

Span-ASTE mastered explicit span reachability, but the inability to emit NULL tuples forced a paradigm shift to generative seq2seq.

The Bridge from Span-ASTE to Generative ByT5

Why end-to-end autoregressive generation became the necessary next horizon.

What Span-ASTE Accomplished

1. Conquered the BIO segmentation ceiling (81.91% reachability).
2. Proved deep biaffine interaction outperforms concatenation.
3. Demonstrated O(1) prefix mean pooling captures multi-word semantics.
4. Produced the project's highest explicit triplet F1 (30.40%).

Why We Pivoted to ByT5

1. 42% of quadruples are implicit and can never be sliced as spans.
2. ACOS requires 4 fields simultaneously, including abstract categories.
3. An autoregressive sequence-to-sequence model can generate literal `NULL` strings.
4. Unifies explicit and implicit extraction into a single model.

Span-ASTE represents the pinnacle of extractive modeling in this repository, establishing the baseline that generative ByT5 was built to surpass.

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